

JOHN ANDREI MERONG MARTINEZ

San Jose Del Monte, Bulacan, Philippines | 0993-446-6975 | martinezjandrei8425@gmail.com
linkedin.com/in/john-andrei-martinez-499a0b343 | github.com/Zeref538 | johnandreimartinez.com

SUMMARY

AI/ML engineering student who ships end-to-end systems: agentic AI, retrieval-augmented generation (RAG), LLM fine-tuning, forecasting and computer vision, taken from dataset to deployed service. Twenty personal projects, eighteen with live demos. Dean's Lister 2023-2026.

SKILLS

Languages: Python, SQL, JavaScript, Java, C, C++

AI/ML: PyTorch, scikit-learn, LLM Fine-Tuning (LoRA, QLoRA, Unsloth), RAG, LangChain, Agentic AI, Ollama, Transformers, CNNs, Computer Vision, Time-Series Forecasting, Model Evaluation

Data: pandas, NumPy, DuckDB, PostgreSQL, MongoDB, SQLite, MySQL, Tableau, Power BI

Backend & Tools: FastAPI, Node.js, React, Docker, Git/GitHub, GitHub Actions, ONNX, n8n, Kaggle

Cloud: Azure OpenAI, Google Cloud, Oracle Cloud, AWS, Vercel, Supabase, Firebase

PROJECTS

YODA - Your Offline Data Agent

2026

Tech: Python, Ollama, FastAPI, pandas, Agentic AI

- Built a local-first data agent that answers questions over CSV and Excel files with no data leaving the machine, running a local Ollama model behind a FastAPI service.
- Designed a tool-calling loop that plans pandas operations, executes them, and validates its own output before answering.

Aegix AI - Contract Compliance Screening

2026

Tech: Python, LangChain, Azure OpenAI, Vector Search, RAG, FastAPI

- Shipped a retrieval-augmented screening tool that flags non-compliant contract clauses and cites the source paragraph behind every flag.
- Built the vector-search retrieval layer over Azure OpenAI embeddings and exposed it as a FastAPI service with a React front end.

Refusal Calibration - LLM Fine-Tuning

2025

Tech: PyTorch, LoRA, QLoRA, Model Evaluation, Python

- Fine-tuned a base model with LoRA so it refuses out-of-scope questions instead of answering them confidently.
- Built a held-out evaluation set that measures over-refusal as well as refusal, so the model is not rewarded for simply declining everything.

APAW - Dam Level and Spill-Risk Nowcaster

2025

Tech: Python, River, Online Learning, Time-Series, GitHub Actions

- Deployed a self-updating forecaster that retrains on new Open-Meteo readings through a scheduled GitHub Actions job, running unattended.
- Used online learning so the model adapts to seasonal shift without a full retrain from scratch.

FORGE - Fake Or Real Image Examiner

2025

Tech: PyTorch, ONNX, Computer Vision, GitHub Actions, JavaScript

- Trained an image classifier that separates AI-generated images from photographs, exported to ONNX so it runs in the browser with no server.
- Automated retraining and redeployment through GitHub Actions.

Which Page Do You Fix First? - Refresh Priority on Search Data

2026

Tech: scikit-learn, DuckDB, pandas, Causal Inference, Python

- Ranked which pages to refresh first from real search-console data, reaching 0.88 Precision@50 against a 0.86 recency baseline.
- Audited label leakage and used bootstrap confidence intervals to show that the initial result was a tie with the baseline.

EXPERIENCE

Backend AI Engineering Intern - FlyRank AI

2026 - Present

- Built backend APIs with database persistence and authentication, plus background jobs and an LLM endpoint across nine assignments.
- Developed capstones for usage metering and Stripe test-mode billing, and semantic image matching with a guard against mismatched images.

ML Engineering Intern - FlyRank AI

2026 - Present

- Prepare and validate datasets, support model testing, and automate workflow steps in Python.
- Build data pipelines and internal automation tools that cut manual handling out of the team's day-to-day work.

Virtual Assistant - Freelance

2025 - 2026

- Ran end-to-end recruitment, data administration and digital record keeping, improving hiring turnaround and record accuracy.

EDUCATION

BS Computer Science, specializing in AI/ML Engineering - Our Lady of Fatima University

2023 - Present

- Dean's Lister 2023-2026, with placements in Python, Database Design and C Programming.

CERTIFICATIONS

Google Advanced Data Analytics Professional Certificate

July 2026

*Coursera***Google AI Professional Certificate**

June 2026

*Coursera***AWS AI Practitioner**

June 2026

*Udemy***Building with the Claude API**

2026

*Anthropic***Introduction to Model Context Protocol**

2026

*Anthropic***Oracle Cloud Infrastructure Certified AI Foundations Associate**

July 2025

*Oracle University***EXTRACURRICULAR ACTIVITIES**

Technical Team Member, English Society. Directorate Member, Junior Philippine Computer Society (JPCS).